

Skin1004 - Data Analyst Intern Portfolio

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1 Exploratory Data Analysis (EDA)

To understand consumer engagement and product performance in the beauty market, we conducted an exploratory data analysis (EDA) using two primary datasets: `cleaned_makeup_reviews.csv` [2] and `cleaned_makeup_products.csv`. The reviews dataset contains user ratings, text, timestamps, and helpfulness counts, while the product dataset includes brand, category, and price information. Together, they reveal how satisfaction and engagement vary across product types and time, forming the basis for subsequent clustering and sentiment modeling.

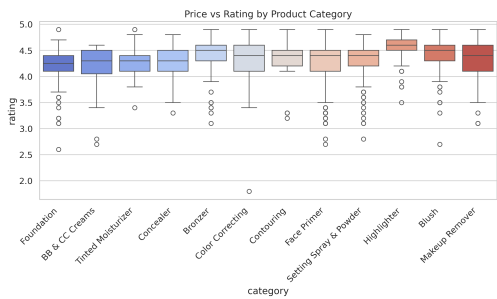


Figure 1: Customer ratings by product category.

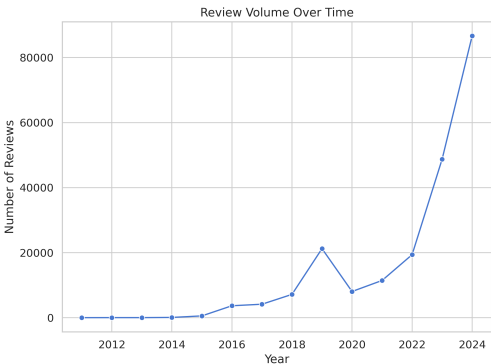


Figure 2: Review volume trend (2010–2025).

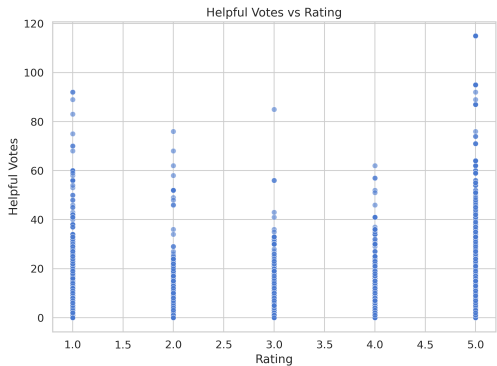


Figure 3: Helpful votes vs. ratings.

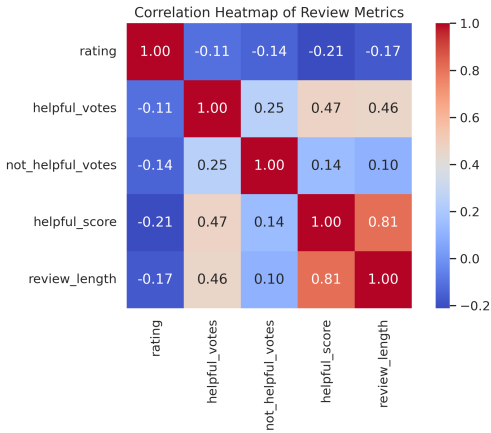


Figure 4: Correlation among engagement metrics.

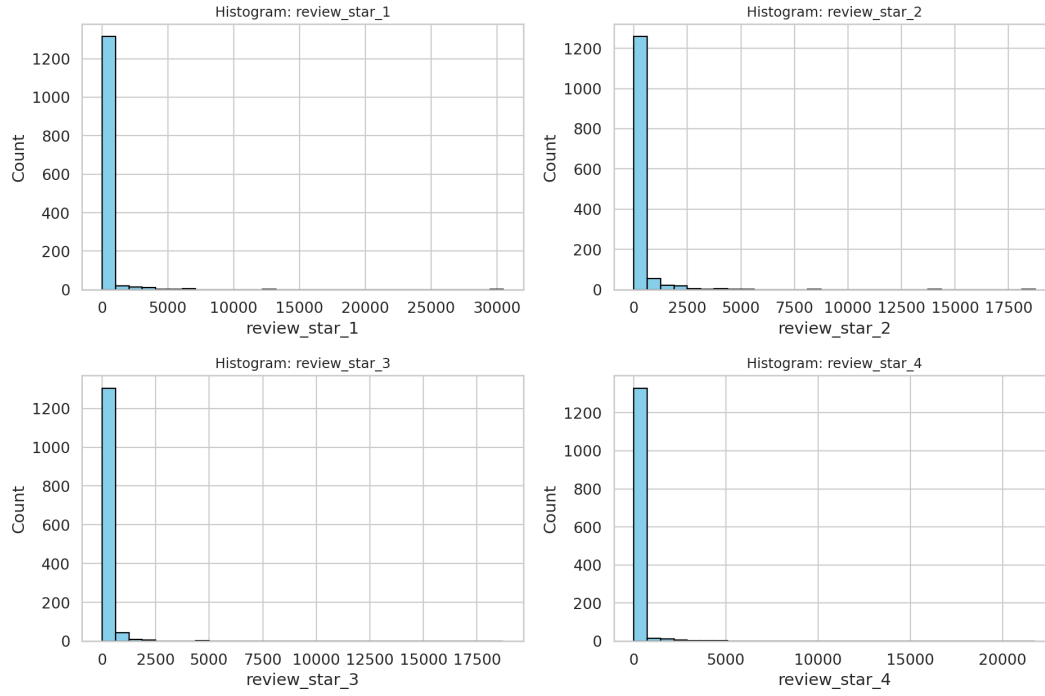


Figure 7: Distributions of review star counts (1-4 stars).

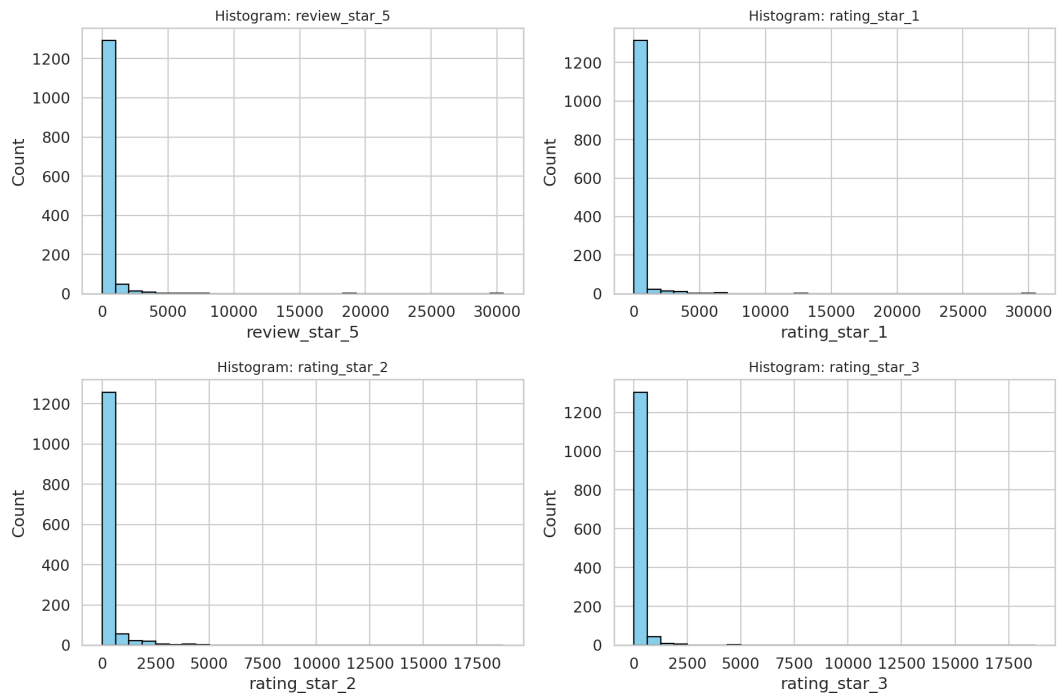


Figure 8: Distributions of 5-star review counts and partial star metrics.

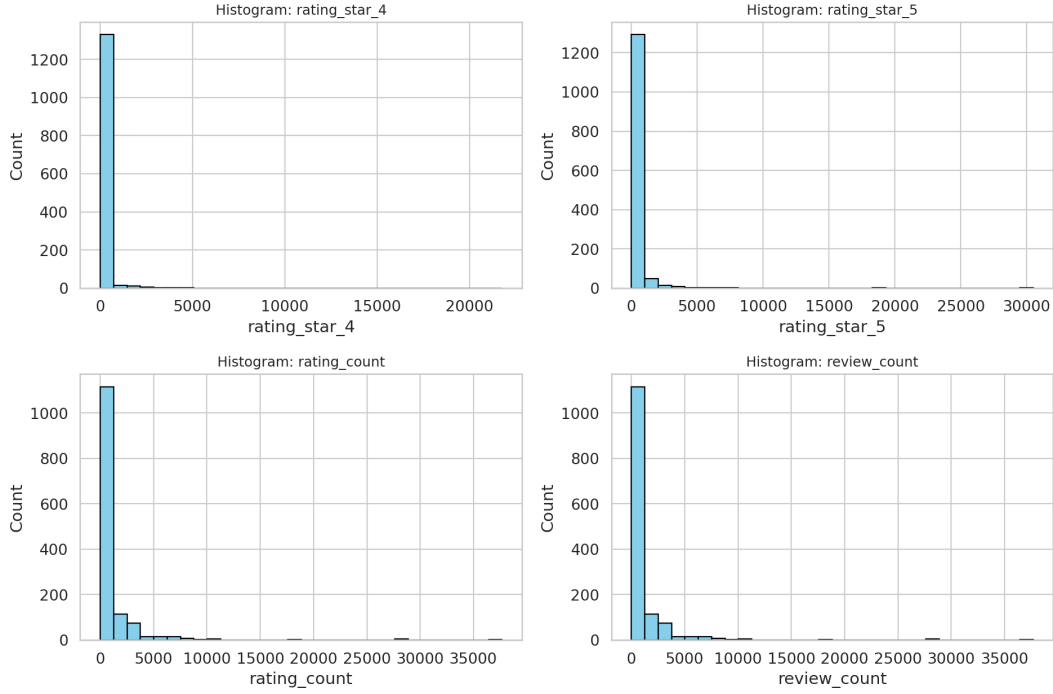


Figure 9: Distributions of rating counts and review counts per product.

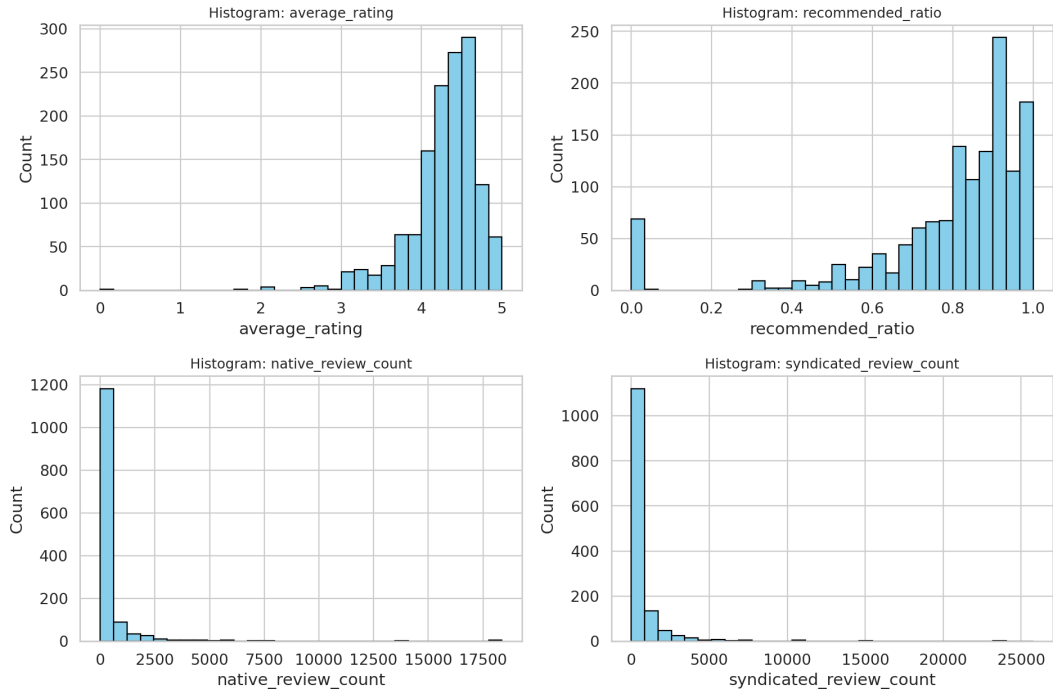


Figure 10: Distributions of average rating, recommendation ratio, and native vs. syndicated review counts.

Overall, prices and shade counts vary widely, while ratings are highly skewed toward the positive end.

Review activity is similarly right-skewed—indicating that a few flagship products dominate engagement. These findings justify applying clustering to identify distinct product and consumer groups within the dataset.

2 Clustering

2.1 Data preprocessing

Before applying any clustering or modeling, several preprocessing steps were done to clean and prepare the numeric data for analysis. The main goals were to handle missing values, fix skewed distributions, and make sure all features were on the same scale.

Handling missing and constant values. Missing values in the numeric dataset were filled using the median of each column. This approach helped keep the distributions consistent without being heavily affected by outliers. Columns that contained only a single unique value (no variation) were removed since they do not contribute to clustering.

Log-transforming skewed variables. Some variables, especially those related to reviews (e.g., `num_reviews`, `review_star_1{5}`, and `rating_star_1{5}`), were extremely right-skewed because of a few products with very large numbers of reviews. To reduce this skew and make the data more balanced, a log transformation was applied using `np.log1p()`, which safely handles zeros.

Standardizing features. After the log transformation, all numeric columns were standardized using `StandardScaler` so that each feature had a mean of 0 and a standard deviation of 1. This step ensured that variables measured on different scales (for example, price vs. review count) contributed equally when computing distances in K-Means, Hierarchical, and K-Medoids clustering.

2.2 K-Means

After preprocessing, the K-Means algorithm was applied to group products based on their numeric features such as price, average rating, and review-related variables. To determine the optimal number of clusters (k), both the Elbow Method and Silhouette Analysis were performed.

Elbow Method. The Elbow Method was used to examine how the within-cluster sum of squares (inertia) changes as k increases. As shown in Figure 11, inertia decreases sharply between $k = 2$ and $k = 4$, after which the improvement becomes more gradual. The point of diminishing returns (“the elbow”) around $k = 4$ suggests that four clusters offer a good balance between simplicity and explanatory power.

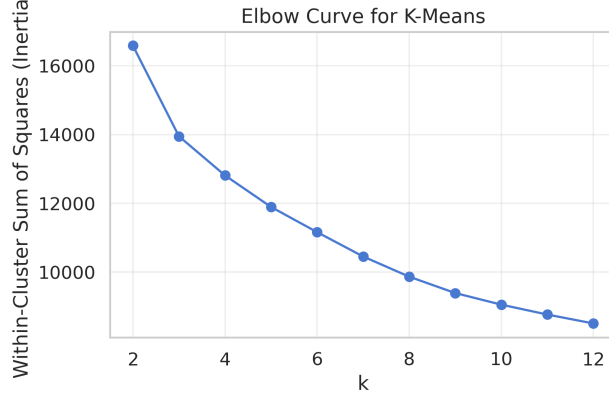


Figure 11: Elbow curve for K-Means showing a clear bend at $k = 4$.

Silhouette Analysis. To validate this choice, Silhouette Scores were calculated for the same range of k values. As illustrated in Figure 12, the highest Silhouette Score occurs at $k = 2$, but the scores remain fairly stable between $k = 3$ and $k = 4$. This stability indicates that $k = 4$ achieves a good separation between clusters while maintaining interpretability.

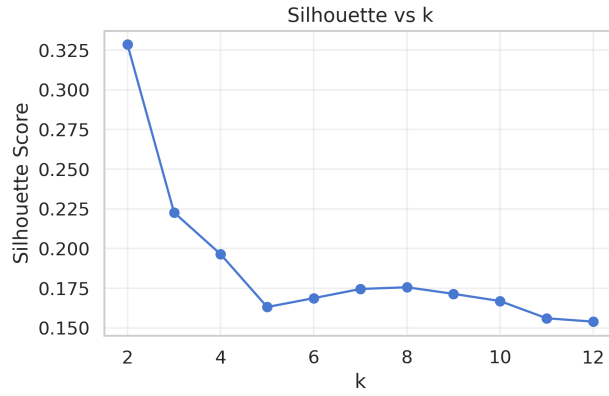


Figure 12: Silhouette scores for different cluster numbers (k), showing consistency near $k = 4$.

Final K-Means Model. Based on the Elbow and Silhouette results, $k = 4$ was chosen for the final model. The K-Means algorithm was initialized 50 times (`n_init=50`) to ensure stability and reproducibility. Each product in the dataset was then assigned to one of the four clusters.

Cluster sizes were as follows:

Cluster	Count
0	146
1	582
2	312
3	333

This four-cluster configuration provided meaningful groupings of products, balancing data compactness and interpretability for further analysis.

PCA 2D Visualization - K-means ($k=4$). To visualize the four clusters generated by the K-Means model, Principal Component Analysis (PCA) was applied to project the standardized feature space into two dimensions. As shown in Figure 13, the clusters appear relatively well separated along the first two principal components. The explained variance ratios for PC1 and PC2 were 0.577 and 0.110, respectively, together accounting for approximately 69% of the total variance.

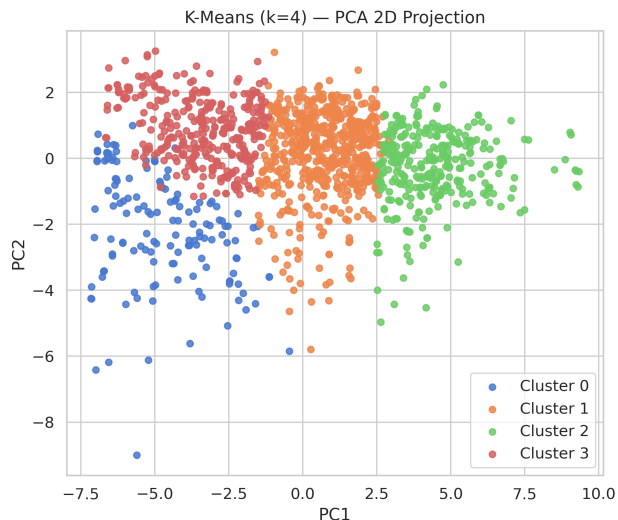


Figure 13: PCA 2D projection of K-Means clusters ($k = 4$).

Silhouette Distribution - K-Means ($k=4$). Figure 14 displays the silhouette scores of each cluster for $k = 4$. The average silhouette score of approximately 0.20 suggests a moderate but interpretable cluster structure. While most samples show positive silhouette values, some observations close to zero indicate boundary cases where samples lie near cluster edges.

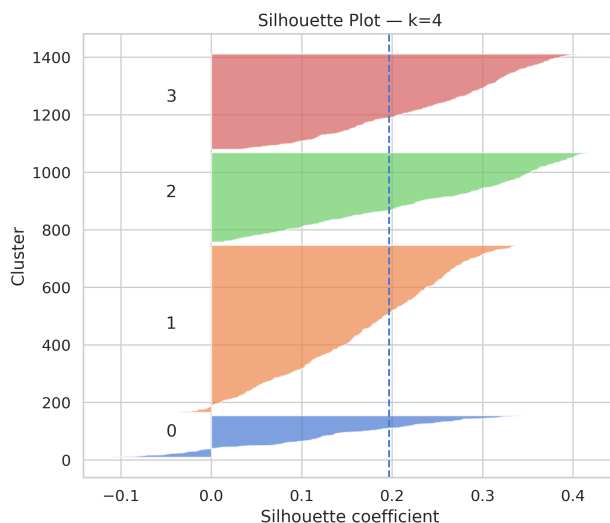


Figure 14: Silhouette plot for $k = 4$, showing within-cluster cohesion and separation (dashed line: overall average).

Cluster Centers and Feature Profiles. The K-Means centroids were computed in standardized space and then inverse-transformed for interpretability. Two adjustments were made: (1) scaling was reversed to restore each variable’s original range, and (2) features that had undergone $\log(1 + x)$ transformation were exponentiated using $\exp(x) - 1$ to recover their natural units. This produced two summary tables (`centers_logspace.csv` and `centers_original_units.csv`) that describe each cluster’s representative profile across all numeric features.

Cluster Center Comparison After identifying four optimal clusters through the K-Means algorithm, we analyzed each cluster’s centroid to interpret their defining characteristics. Figures 15–17 display the per-feature mean values of all clusters, grouped into compact 3×3 grids for clarity.

Cluster 2 consistently exhibits the highest review engagement (e.g., `num_reviews`, `rating_count`) and total ratings across all star categories, indicating a group of highly popular products. Cluster 0 represents lower-priced and less-reviewed products, while Clusters 1 and 3 contain moderately priced or premium items with balanced review distributions and relatively higher average ratings.

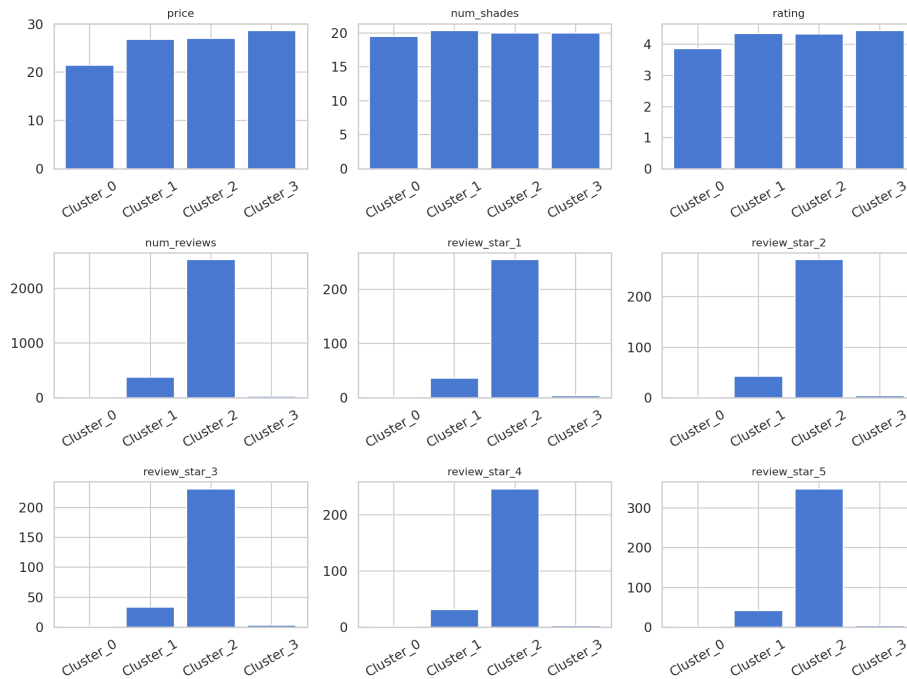


Figure 15: Cluster center comparisons (Part 1): Price, number of shades, rating, and detailed star-level review metrics.

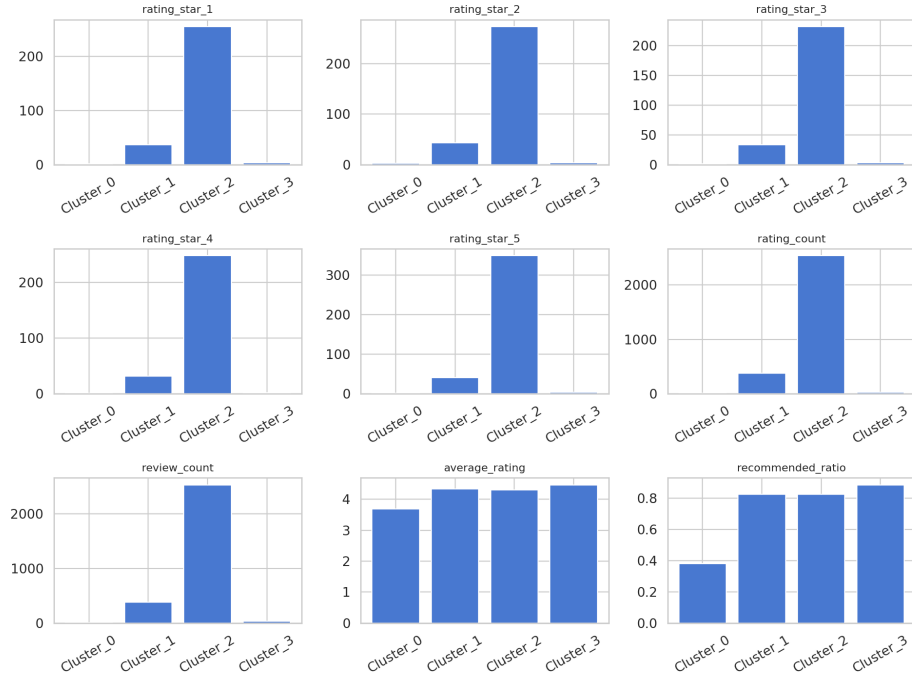


Figure 16: Cluster center comparisons (Part 2): Star-level rating counts, review volume, and satisfaction-related metrics.

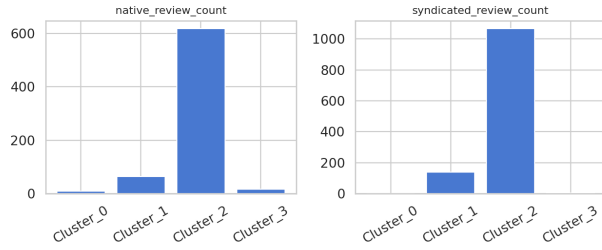


Figure 17: Cluster center comparisons (Part 3): Final review distribution and recommendation ratio.

3D PCA Visualization of K-Means Results To provide a more comprehensive visualization of the cluster separation, a three-dimensional PCA projection was generated using the first three principal components. This projection captures over 70% of the total variance and highlights clear cluster boundaries in reduced space.

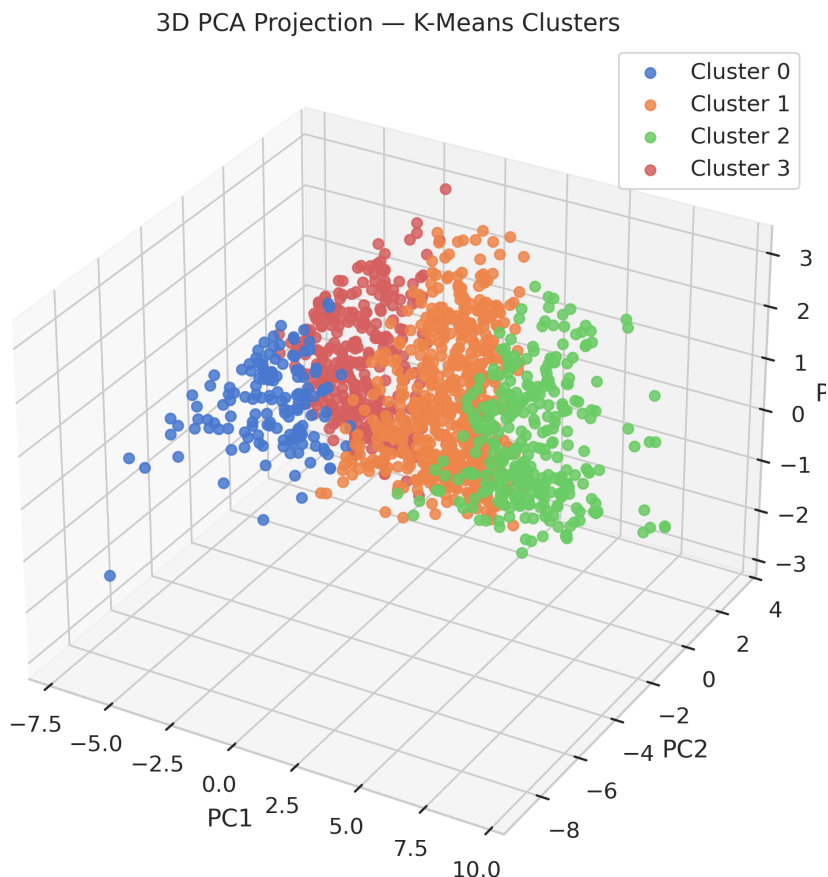


Figure 18: 3D PCA Projection of the K-Means clusters ($k = 4$).

The 3D PCA view provides an intuitive perspective on how the K-Means algorithm effectively partitions products with similar attributes into distinct groups, aligning with the statistical validation results above.

2.3 Hierarchical Clustering

Hierarchical agglomerative clustering was implemented using the Ward linkage method through the `scipy.cluster.hierarchy` and `sklearn.cluster` libraries. This algorithm begins by treating each product as an individual cluster and iteratively merges the two closest clusters based on the minimum variance criterion, resulting in a nested tree structure that reflects the relationships between data points. The Ward method was chosen because it minimizes within-cluster variance at each merge step, making it well suited for continuous, standardized variables such as price, average rating, and review counts.

To ensure comparability across features, the same standardized dataset used in K-Means clustering was applied here. The distance matrix was computed using Euclidean distance, and the full linkage process was visualized using dendrograms to identify natural separations in the data. The implementation followed these steps:

1. Preprocessed features were scaled with `StandardScaler` to achieve zero mean and unit variance.
2. Pairwise distances were calculated using Euclidean distance.
3. The Ward linkage criterion was applied to iteratively merge similar data points.
4. Dendrograms were plotted to visualize merge distances and evaluate cluster stability.
5. Silhouette analysis was performed across $k = 2$ to 12 to determine the optimal cluster count.

This implementation provides a hierarchical representation of the beauty product landscape, capturing relationships at multiple levels of granularity. Compared to K-Means, this approach emphasizes interpretability by revealing how smaller product groups progressively merge into larger, more generalized segments.

Determining the Number of Clusters. To identify an appropriate cluster count, we first examined the last 50 merge distances (Figure 19). A sharp increase in distance toward the end of the curve suggests a natural separation in the data. This pattern is reinforced by the truncated dendrogram (Figure 20), which displays four major cluster branches before a significant jump in linkage distance. These findings point to $k = 4$ as a suitable configuration, consistent with the K-Means results.

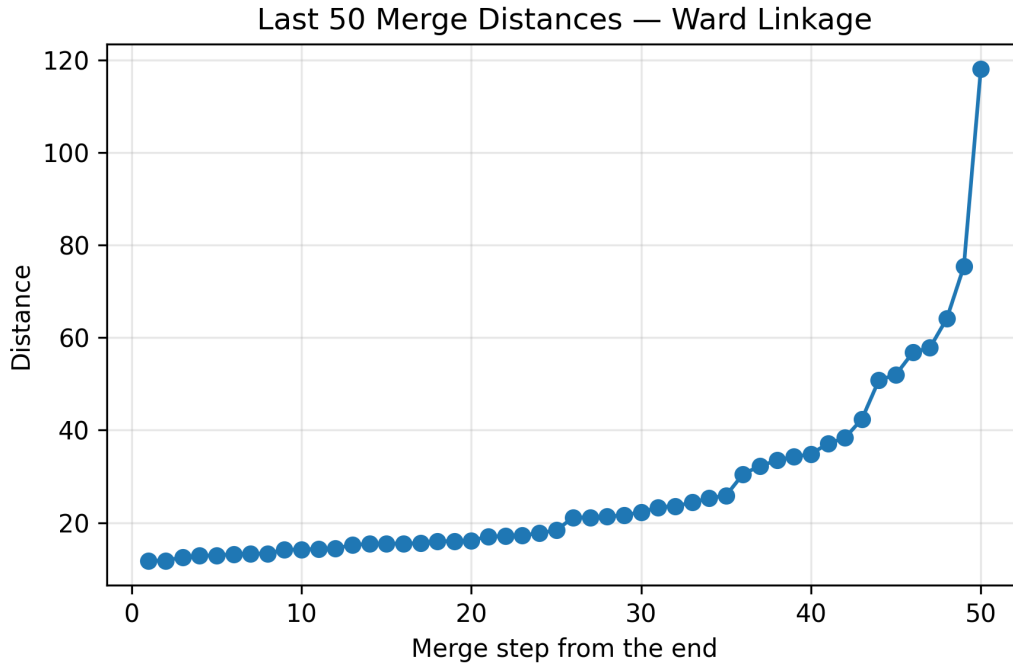


Figure 19: Last 50 merge distances using Ward linkage, highlighting a sharp rise near the final merges.

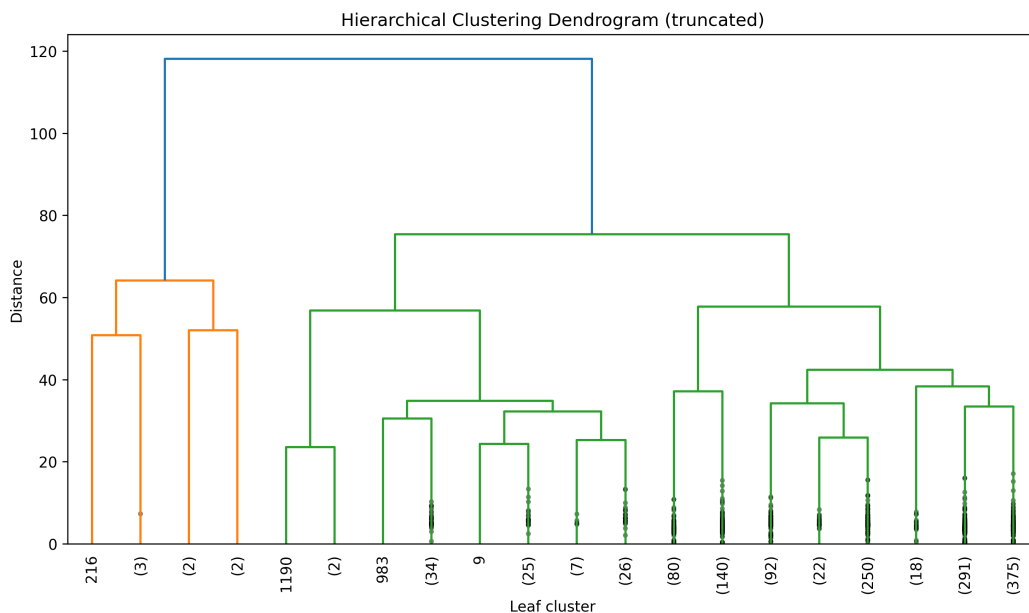


Figure 20: Truncated hierarchical dendrogram showing the major cluster separations ($k = 4$).

Silhouette Analysis. To validate the choice of four clusters, silhouette scores were computed for a range of k values (Figure 21). The average silhouette coefficient peaks at $k = 2$ but remains locally stable around $k = 4$, suggesting a balance between cohesion and separation. The detailed silhouette plot for $k = 4$ (Figure 22) shows that while most clusters maintain positive silhouette values, some overlap exists between adjacent groups, reflecting the gradual structure typical of hierarchical clustering.

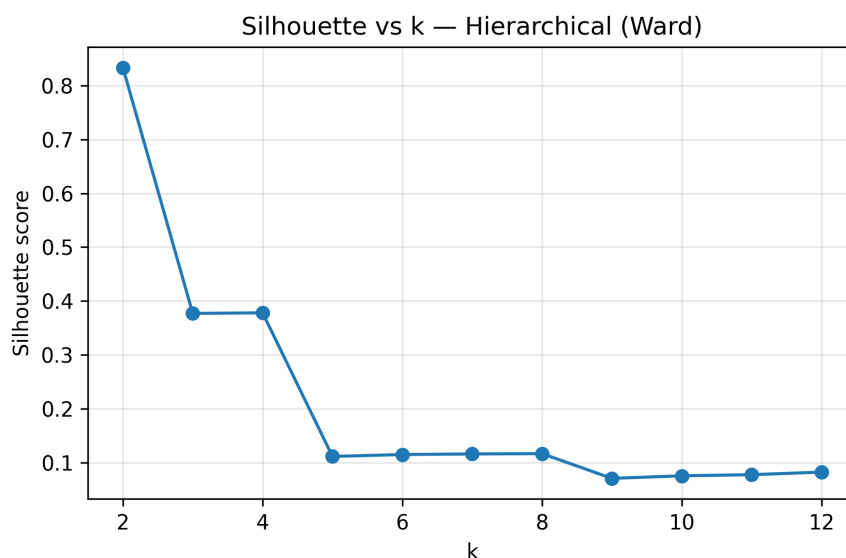


Figure 21: Silhouette scores for different cluster counts (k), showing local stability near $k = 4$.

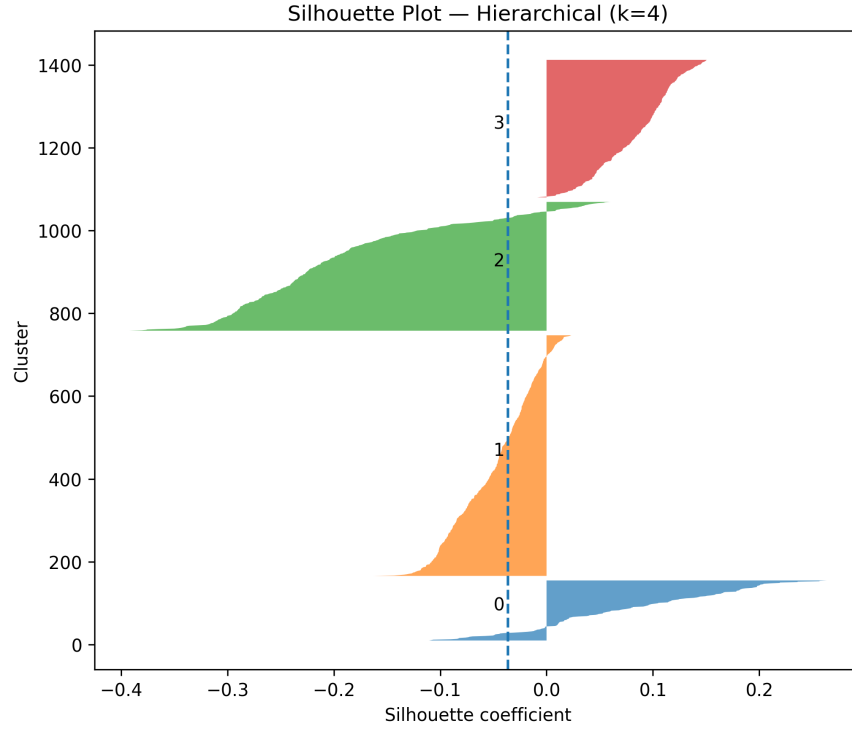


Figure 22: Silhouette plot for hierarchical clustering ($k = 4$). The dashed line indicates the average silhouette score.

Cluster Visualization. To visualize how the hierarchical clusters are distributed in feature space, a PCA projection was performed (Figure 23). The first two principal components capture the majority of variance in the dataset and reveal moderate separation among the four clusters, consistent with the K-Means results. Additionally, a full dendrogram based on a random 400-sample subset (Figure 24) provides a global view of hierarchical relationships, illustrating how smaller product groups merge into broader segments.

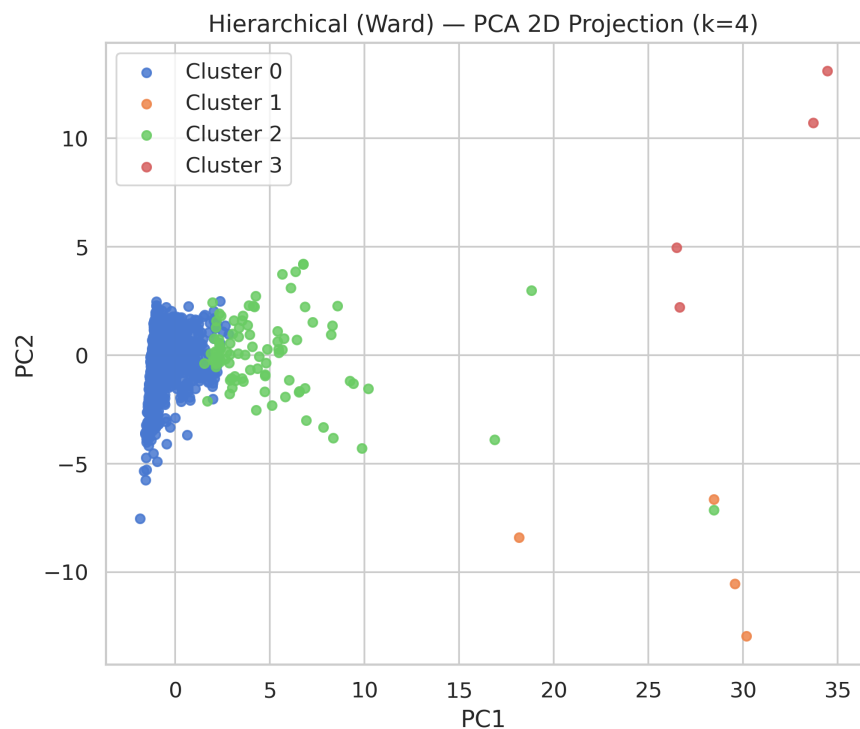


Figure 23: PCA 2D projection of hierarchical clusters ($k = 4$) using Ward linkage.

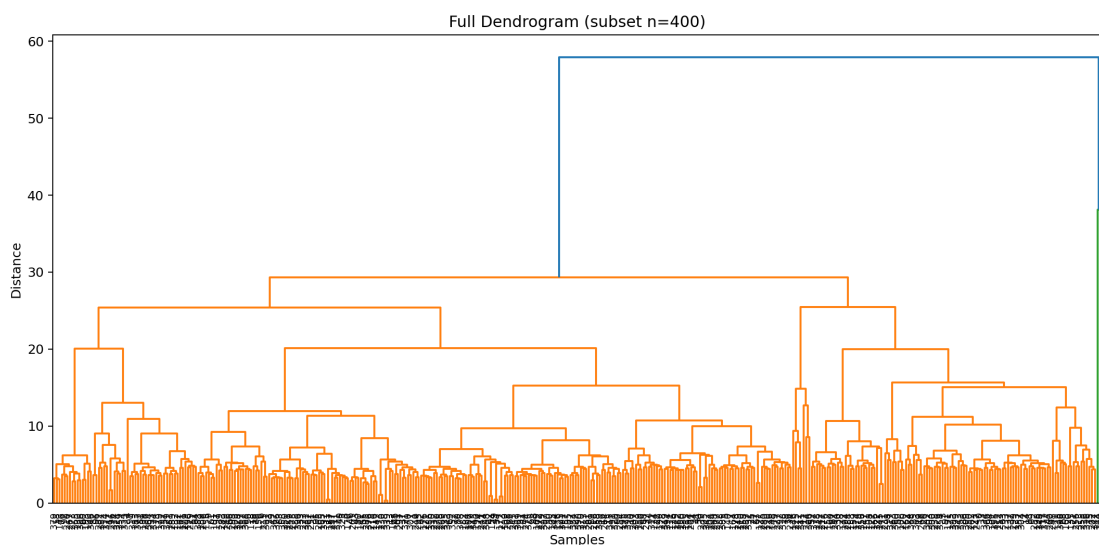


Figure 24: Full dendrogram on a random subset ($n = 400$) illustrating the nested structure of cluster merges.

Interpretation. Overall, hierarchical clustering confirms the presence of four major product groups within the dataset, aligning with K-Means results but offering a more interpretable merge structure. This reinforces the robustness of the segmentation and provides a complementary perspective on how products of varying price, popularity, and rating distribution relate to one another across different levels of granularity.

2.4 K-Medoids (PAM)

To complement K-Means and Hierarchical clustering, we implemented Partitioning Around Medoids (PAM) using the `pyclustering` library. Unlike K-Means, which uses centroids that may fall outside the data manifold, K-Medoids selects *actual* observations (medoids) as cluster representatives, improving robustness to outliers and offering interpretable exemplars.

Implementation. We standardized the same numeric feature set used in prior sections (`StandardScaler`) and ran PAM for $k \in \{2, \dots, 12\}$ with multiple random initializations per k (Algorithm: `kmedoids` with Euclidean distance). For each k , we retained the run with the highest silhouette score. We recorded (i) the total sum of distances to medoids (*Elbow*) and (ii) the average silhouette coefficient for model selection.

Model selection. Figure 25 shows a diminishing return in cost as k increases, and Figure 26 plots silhouette scores by k , which peak at $k=2$ but remain locally stable around $k=4$. Balancing compactness, separation, and interpretability with our downstream profiling, we select $k=4$.

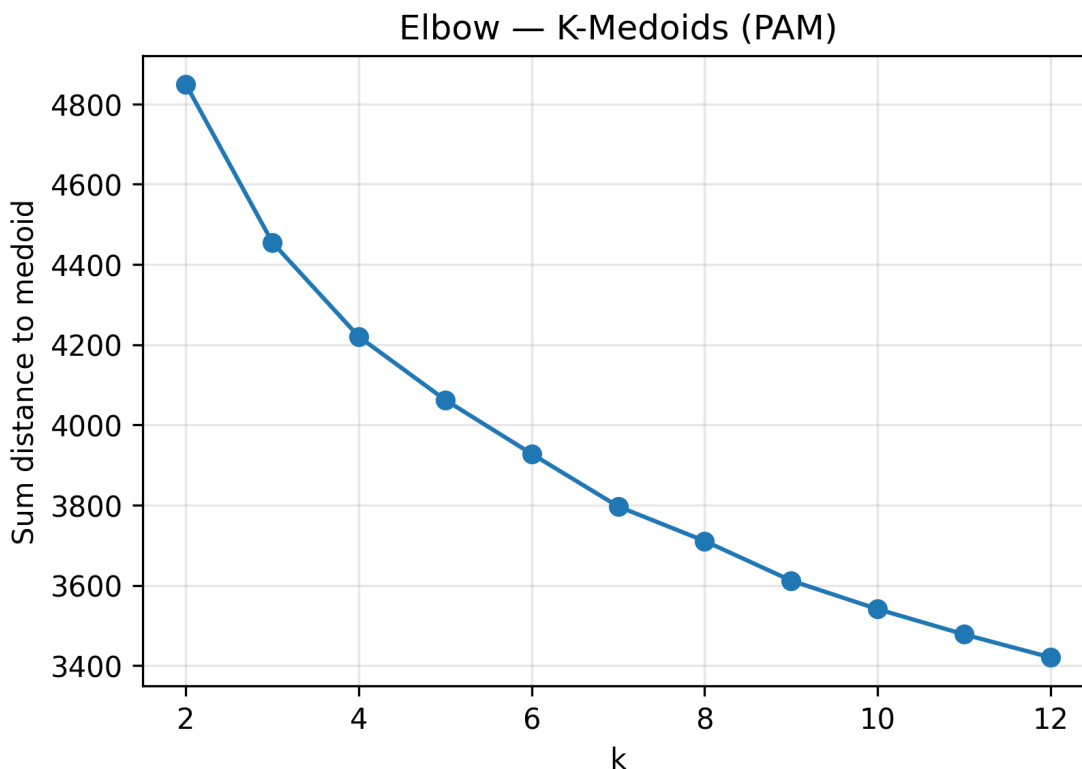


Figure 25: Elbow — K-Medoids (PAM): sum of distances to each cluster medoid vs. k .

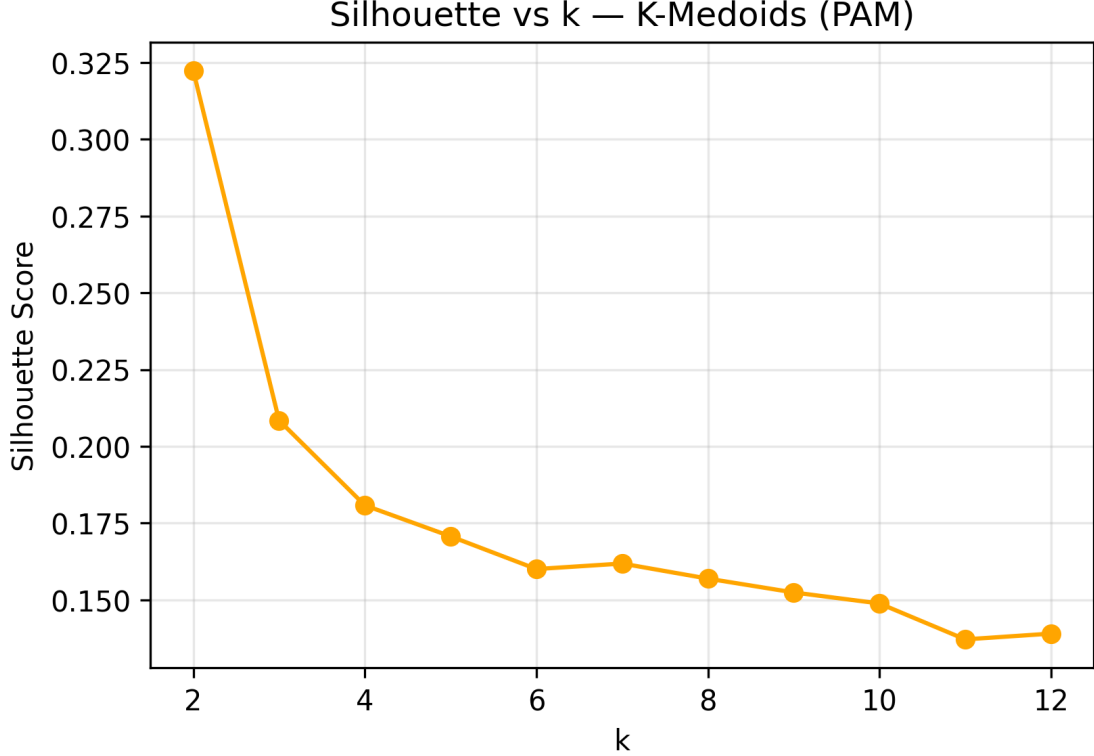


Figure 26: Silhouette vs. k — K-Medoids (PAM). Peak at $k=2$ and local stability near $k=4$.

Cluster assignment and medoids. With $k=4$, PAM assigns each product to the nearest medoid (an in-dataset representative). Table 1 reports cluster sizes; Table 2 lists the medoid products for transparency and interpretability.

Table 1: Cluster sizes for K-Medoids ($k=4$).

Cluster	0	1	2	3
Count	275	384	434	280

Table 2: Medoid products selected by PAM.

Product name	Brand	Price	Avg. rating
Liquid Lip Blush & Cheek Tint	Benefit Cosmetics	26.0	4.11
Shape Tape Radiant Medium Coverage Concealer	Tarte	28.0	4.28
Mini Translucent Loose Setting Powder Ultra-Blur	(brand)	28.0	4.33
Rinse-Off Eye Makeup Remover Solvent	(brand)	26.0	4.49

Low-dimensional view. A PCA projection of the standardized features (PC1 vs. PC2) provides a qualitative check on separation (Figure 27). Clusters form contiguous regions, with partial overlap consistent

with the silhouette distributions.

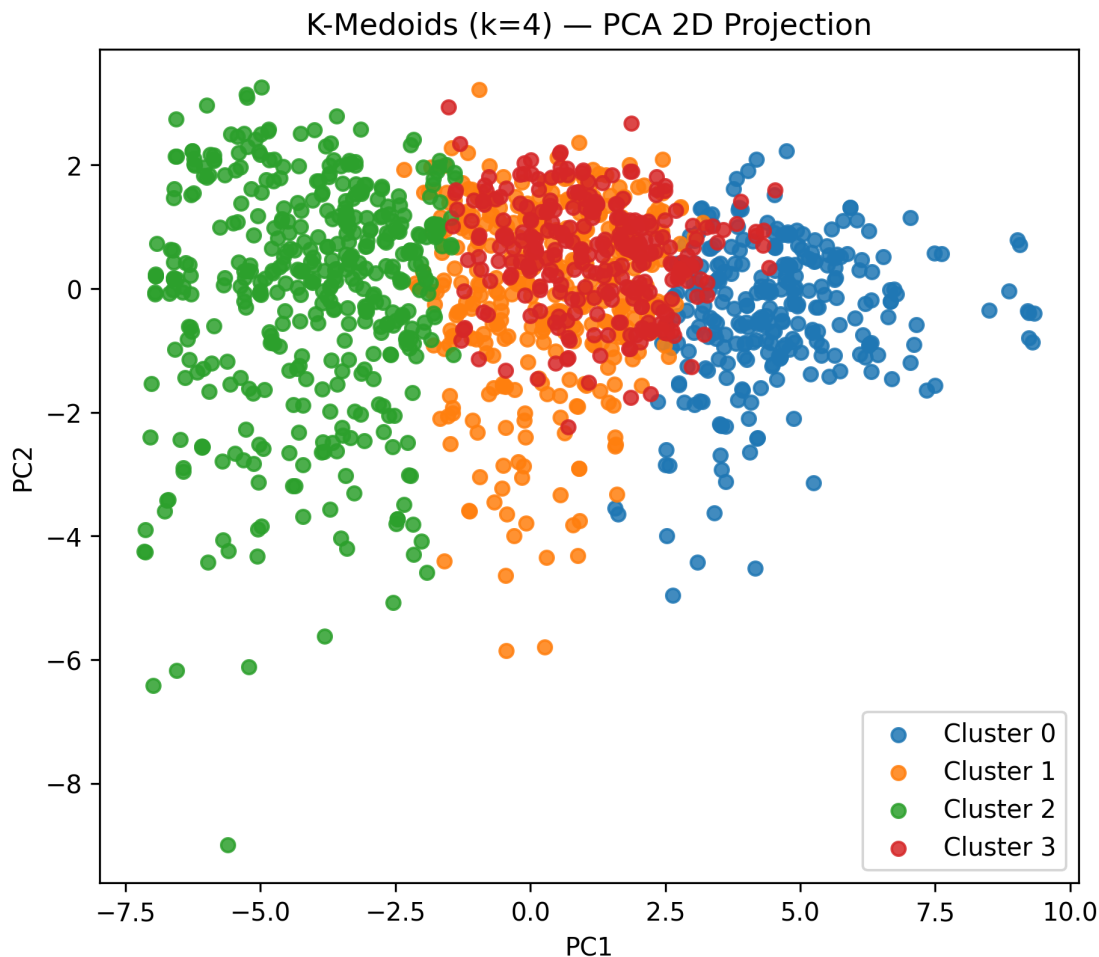


Figure 27: PCA 2D projection — K-Medoids ($k=4$).

Cluster profiles. To interpret segments, we summarize each cluster’s central tendencies across the modeling features. Table 3 reports means; Table 4 reports medians; Table 5 provides interquartile ranges (IQR) as an effect-size view. (These tables can be auto-generated from notebooks and exported as `.tex` via `pandas.DataFrame.to_latex()`.)

Table 3: Per-cluster mean of key features (excerpt).

Feature	C0	C1	C2	C3
price	26.56	26.95	26.42	27.15
num_shades	22.90	22.22	15.40	22.81
rating	4.29	4.29	4.22	4.46
num_reviews	4024.97	396.57	32.40	650.04
review_star_5	1054.50	166.02	5.83	28.80
...	...	...	...	...

Table 4: Per-cluster median of key features (excerpt).

Feature	C0	C1	C2	C3
price	25.0	28.0	24.0	28.5
num_shades	22.0	20.0	7.5	24.0
rating	4.3	4.4	4.3	4.5
num_reviews	2496.0	300.0	22.0	522.0
review_star_3	195.0	33.5	2.0	27.5
...	...	...	...	...

Table 5: Per-cluster IQR (Q3–Q1) of key features (excerpt).

Feature	C0	C1	C2	C3
price	12.25	10.00	9.00	0.75
num_shades	13.81	38.00	38.00	39.00
rating	0.30	0.20	0.30	0.30
native_review_count	30.75	0.0	0.0	2217.5
syndicated_review_count	177.25	–	–	648.00
...	...	...	...	...

Interpretation. K-Medoids identifies four segments that are broadly consistent with prior methods while providing concrete exemplars. Cluster 2 represents low-engagement products with minimal reviews; Cluster 0 concentrates highly popular items with large review volumes; Clusters 1 and 3 capture mid-range or premium products with balanced engagement. Because medoids are real products, they serve as practical anchors for qualitative inspection and downstream recommendation or benchmarking tasks.

3 Sentiment Analysis

3.1 Data Preprocessing and Model Training

To quantify review polarity at scale, we applied a BERT (Bidirectional Encoder Representations from Transformers) sentiment model to review text and mapped outputs to a five-point rating scale. BERT was chosen for its ability to capture context-dependent sentiment, which is especially useful for domain-specific text like makeup reviews. The goal was to validate that predicted sentiment aligns with known labels (positive/negative subsets) and derive brand and product-level aggregates for analysis.

Preprocessing. Text from reviews were converted to lowercase and cleaned to remove punctuation, digits, and redundant words or phrases. Stopwords were removed using the NLTK English list, and additional domain-specific redundancies were filtered to reduce noise. The cleaning function tokenized text to prepare it for model input.

Model and Inference. A BERT sentiment classifier (`nlptown/bert-base-multilingual-uncased-sentiment`) was applied via the `transformers` pipeline. For each review, the model output a label in $1, \dots, 5$, where 1 indicates very negative sentiment and 5 very positive. Ratings ≥ 3 were treated as *predicted positive* and < 3 as *predicted negative*.

Validation. Using held-out subsets with known polarity, we computed accuracy and classification metrics. The BERT sentiment model achieved an accuracy of 74.72 percent and demonstrated reasonably strong alignment with known labels while highlighting that the model is more likely to correctly identify positive reviews.

3.2 BERT Sentiment Model Analysis

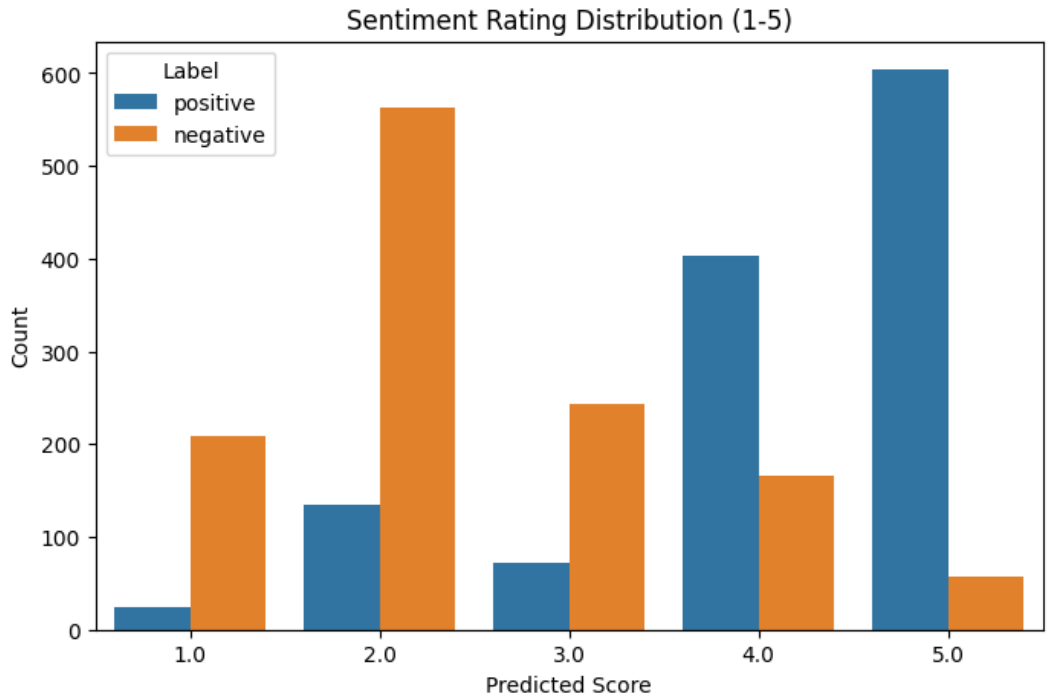


Figure 28: Distribution of predicted 1-5 sentiment scores for reviews with known polarity.

The sentiment rating distribution graph shows how reviews are spread across the 1–5 star scale for positive and negative reviews. Lower scores (1–2) are dominated by negative reviews, with around 200–550 counts compared to much fewer positive reviews, indicating that low ratings mostly reflect negative sentiment. Mid-range scores (3) have moderate counts for both positive and negative reviews, but more reviews are classified with negative sentiment. Higher scores (4–5) are strongly dominated by positive reviews, with 400–600 counts for positive and very few negative, illustrating that high ratings are generally associated with positive sentiment. Predicted scores skew high for truly positive reviews and low for truly negative reviews, indicating alignment between model output and ground-truth subsets, indicating that the sentiment scores mostly align well with the expected polarity of the reviews.

Example Review: This lipstick is amazing! It's very pigmented and long-lasting.
Predicted sentiment: 5 stars, Confidence: 0.93

Example Review: The foundation is patchy and dries my skin.
Predicted sentiment: 2 stars, Confidence: 0.54

Example Review: Okay product, nothing special.
Predicted sentiment: 3 stars, Confidence: 0.85

Figure 29: Predicted quantitative sentiment from model for example reviews.

The sentiment model can be applied to predict the sentiment of new, future reviews and estimate their corresponding ratings based on keywords and estimated context. For example, the first review is predicted as 5 stars with high confidence (0.93). A negative review, “The foundation is patchy and dries my skin,” received a lower prediction of 2 stars with moderate confidence (0.54). Neutral feedback was predicted as 3 stars with high confidence (0.85). These predictions demonstrate the model’s accuracy and how it can provide insights for future customer reviews.

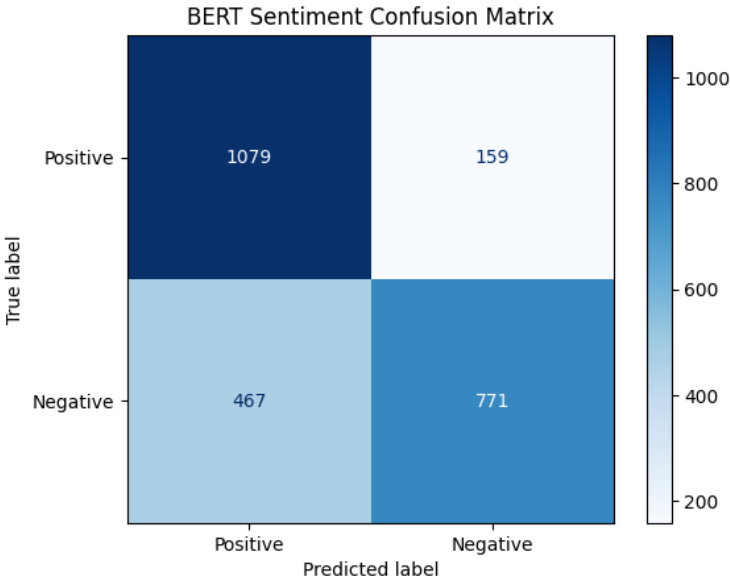


Figure 30: Confusion matrix showing the performance of the sentiment model.

The confusion matrix illustrates the performance of the sentiment prediction model by comparing the true and predicted labels. Of the positive reviews, 1,079 were correctly predicted as positive, while 159 were misclassified as negative. For negative reviews, 771 were correctly identified, but 467 were incorrectly predicted as positive. Overall, the confusion matrix highlights that there were a greater number of correct predictions and the model performs well on positive reviews but has a higher rate of misclassification for negative reviews, providing insight into areas where prediction errors could occur.

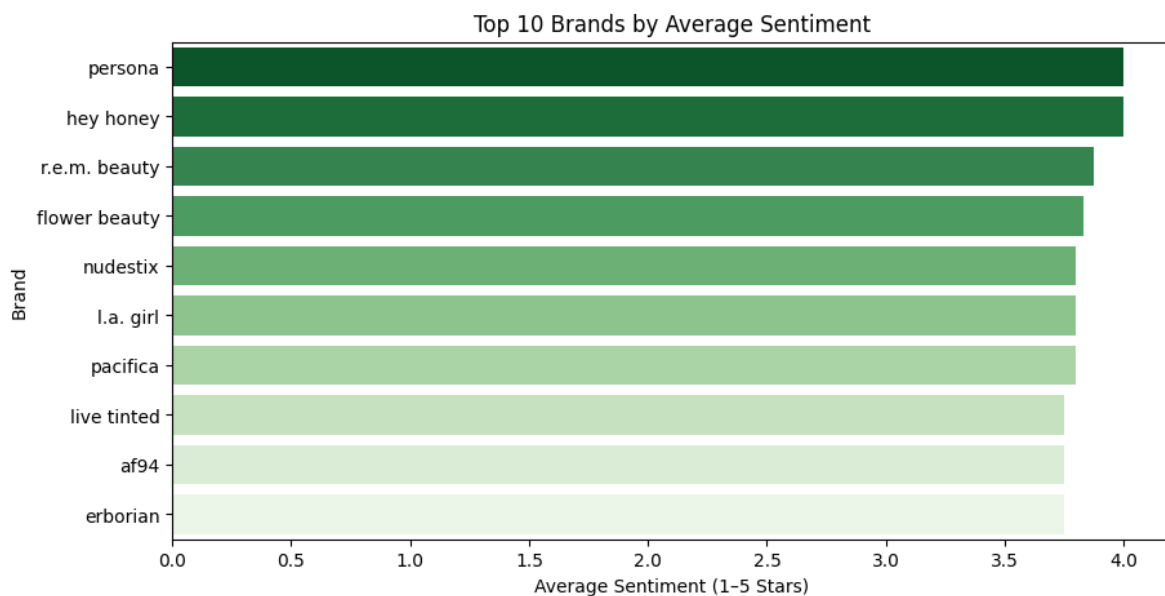


Figure 31: Top 10 brands by average predicted sentiment (1-5).

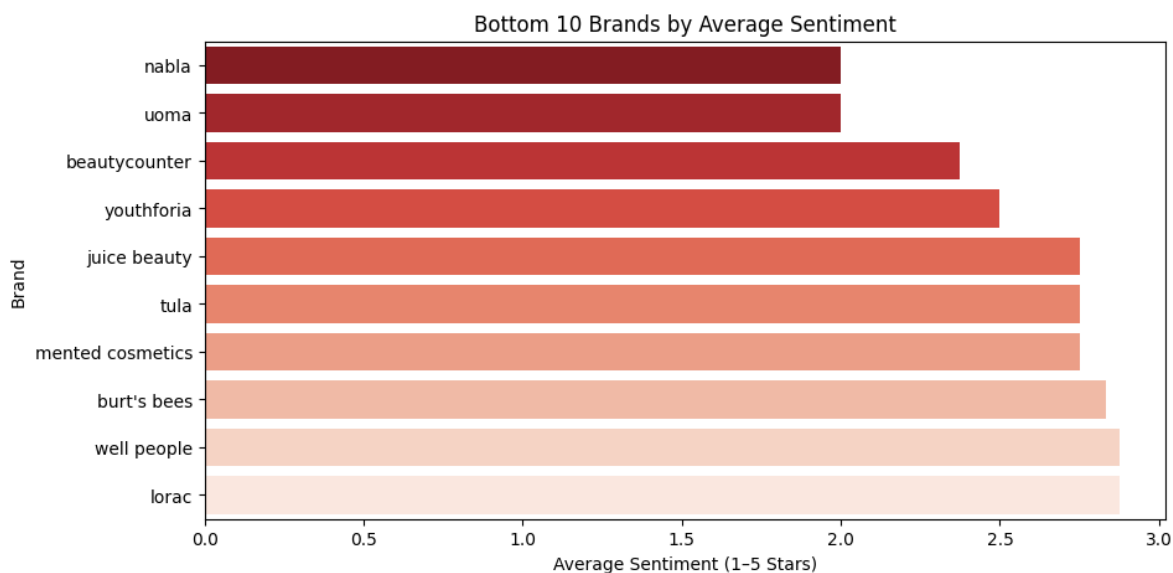


Figure 32: Lowest 10 brands by average predicted sentiment (1-5).

The analysis of average sentiment scores by brand highlights clear differences in customer perception. The top 10 brands—Persona, Hey Honey, R.E.M. Beauty, Flower Beauty, Nudestix, L.A. Girl, Pacifica, Live Tinted, AF94, and Erborian—consistently received higher sentiment scores, indicating strong positive reviews and customer satisfaction. In contrast, the bottom 10 brands—Nabla, Uoma, Beautycounter, Youthforia, JuiceBeauty, Tula, Mented Cosmetics, Burt’s Bees, Well People, and Lorac—show lower sentiment scores, suggesting that customers wrote about more mixed or negative experiences in their product reviews. These insights can help brands understand their relative positioning in the market and identify opportunities to

improve product or marketing strategies for better customer perception.

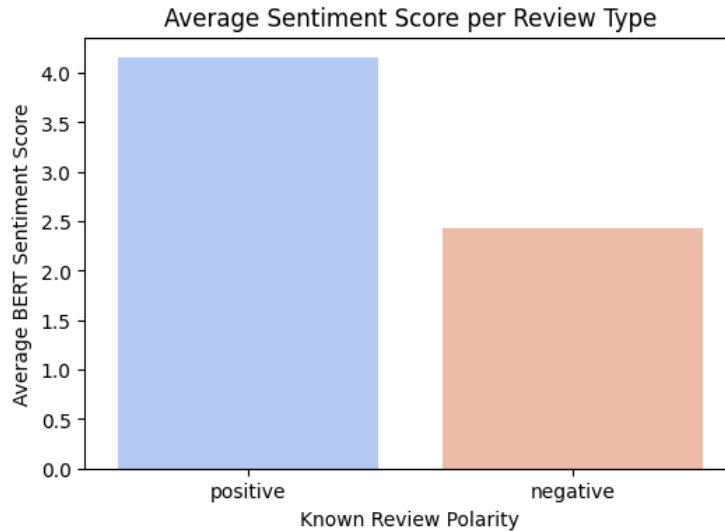


Figure 33: Average predicted sentiment score by review type (positive vs. negative).

The average sentiment score per review type graph shows that reviews labeled as positive have an average sentiment of approximately 4.2, while reviews labeled as negative have an average of about 2.4. This indicates that the model’s predicted scores align well with the expected polarity of the reviews, with an average of higher scores corresponding to positive sentiment and lower scores to negative sentiment.

This also suggests that brands can use sentiment scores as a benchmark for customer satisfaction: aiming for reviews that consistently achieve scores closer to 4–5 indicates strong positive perception, while scores near 2–3 may signal areas needing improvement. By monitoring these average sentiment scores, brands can identify which products or features are well liked by customers and which may require adjustments to improve ratings and customer satisfaction.

Implementation Notes. We used `transformers`, `datasets`, and `torch` for inference; `pandas`/`numpy` for wrangling; and `matplotlib`/`seaborn` for figures. Classifier predictions were evaluated with `accuracy_score` and `classification_report`. To reduce compile overhead in L^AT_EX, all plots were exported as compressed PNGs.

References

- [1] Z. Sarkar. Luxxify: Ulta makeup products. <https://www.kaggle.com/datasets/zusmani/ulta-makeup-products-reviews>, 2024. Kaggle dataset.
- [2] Z. Sarkar. Luxxify: Ulta makeup reviews. <https://www.kaggle.com/datasets/zusmani/ulta-makeup-products-reviews>, 2024. Kaggle dataset.